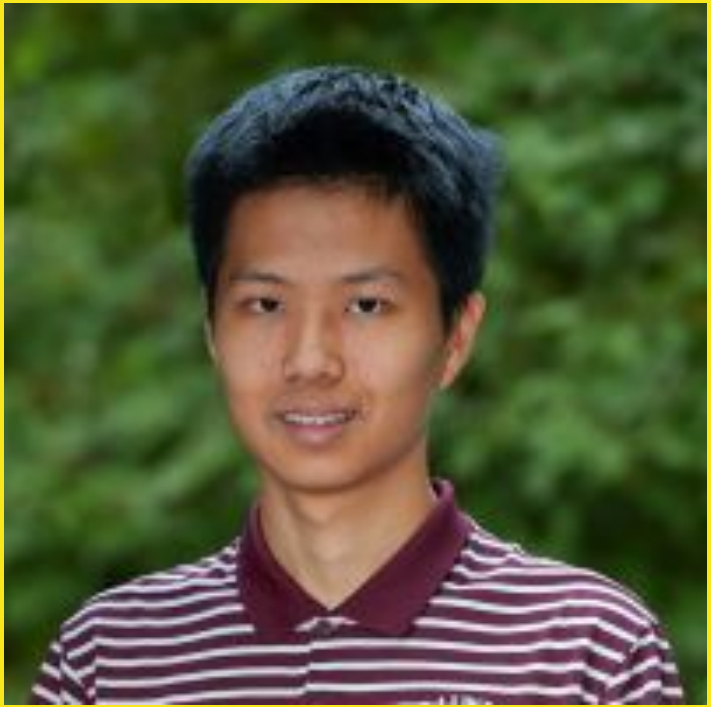
Welcome to the Artificial Intelligence Workshop (Week 1)

Let's Learn How Computers Learn from Information

About Me



Name: Peikang Hu

Current Education: Rising senior at St. Luke's School, New Canaan, Connecticut

Experiences in coding(Python, C++, Java..) and machine learning

Icebreaker

How Many AI Tools Have You Used?

Raise your hand if you've used...

- □ ChatGPT
- □ Siri or Alexa
- □ Google Translate
- □ YouTube recommendations
- □ TikTok or Instagram
- □ Face ID on a phone

Did you know?

You've probably been using AI almost every day!



Workshop Overview

Week 1 – What is Artificial Intelligence (AI)?

Goal: Spark curiosity and understand what AI is.

Week 2 – How Computers Learn

Goal: Introduce linear regression and its application.

Week 3 – A Continuation of Week 2 BUT with more Fun!

Goal: Introduce more advanced AI models.

Week 4 – A Continuation of Week 3 BUT with more Fun!

Goal: Introduce more advanced AI models.

Learning Objectives

By the End of Today, You Will Be Able To...

- ✅ Explain what AI is in your own words.
- ✅ Recognize where AI is used every day.
- ✅ Understand the basic idea of how AI learns.
- ✅ Separate AI facts from AI myths.
- ✅ Be ready to build your first AI model next week.

The History of AI: Before AI Existed

Have Humans Always Dreamed of Intelligent Machines?

- Decades ago, people imagined robots and thinking machines.
- Ancient stories described mechanical servants and artificial beings.
- Science fiction inspired scientists to make these ideas real.



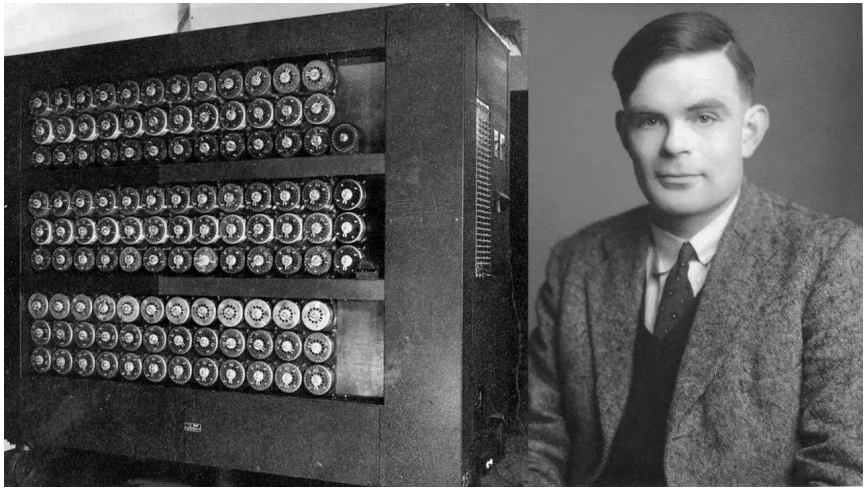
Movie: Metropolis(1927)

The History of AI: The Birth of AI (1950)

Can Machines Think?

- In 1950, mathematician **Alan Turing** asked:

"Can machines think?"



- He created the **Turing Test**, a way to test whether a computer could imitate human conversation.
- This question inspired modern AI research.

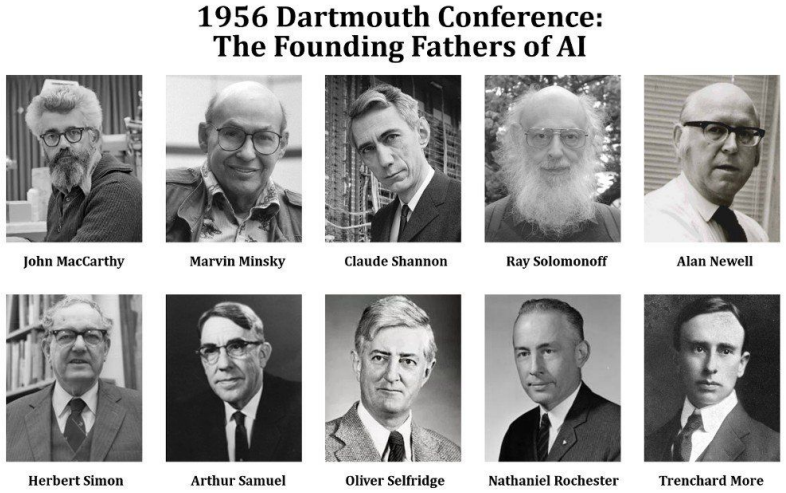
https://www.youtube.com/watch?v=ICO0y8GPKVo

The History of AI: The Name "Artificial Intelligence" (1956)

AI Gets Its Name

- In 1956, scientists met at the Dartmouth Conference.
- They officially named the field **Artificial Intelligence**.
- They believed computers would become very smart within a few decades.

Fun Fact: They were much more optimistic than reality!



The History of AI: Early AI (1960s–1970s)

The First AI Programs

Early AI could:

- Play simple games
- Solve math problems
- Follow logical rules
- Answer basic questions

But...

- Computers were slow.
- There wasn't enough data.
- AI wasn't very powerful yet.



The History of AI: The First AI Winter (1970s–1980s)

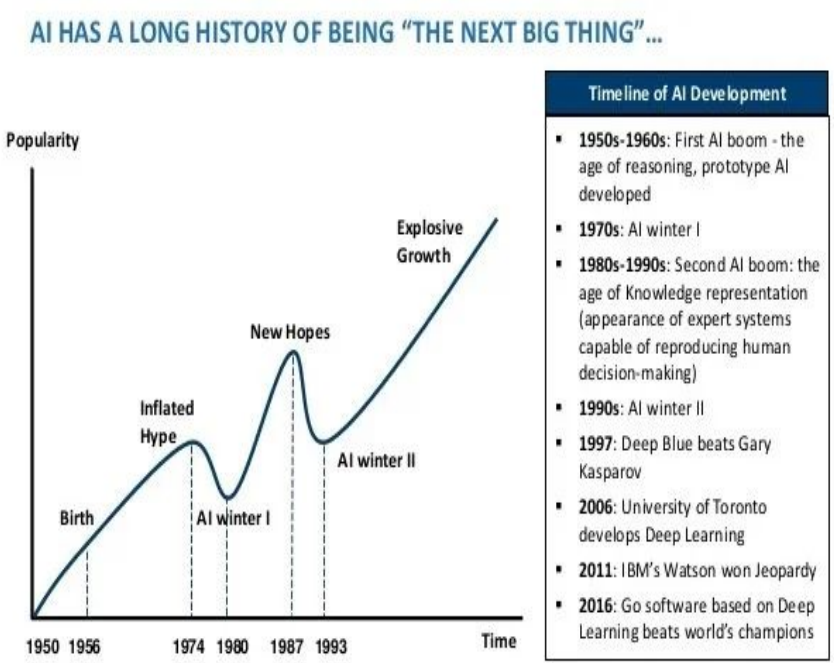
AI Loses Momentum

- Expectations for AI were too high.
- Progress was slower than scientists prec
- Funding and interest decreased.

This period became known as an **AI Winter**.

Key idea:

💡 AI development happens in cycles of excitement and challenges.



The History of AI: AI Defeats a Chess Champion (1997)

1997 — Deep Blue Beats Garry Kasparov

- IBM's **Deep Blue** defeated world chess champion Garry Kasparov.
- It showed that computers could outperform humans in specific tasks.



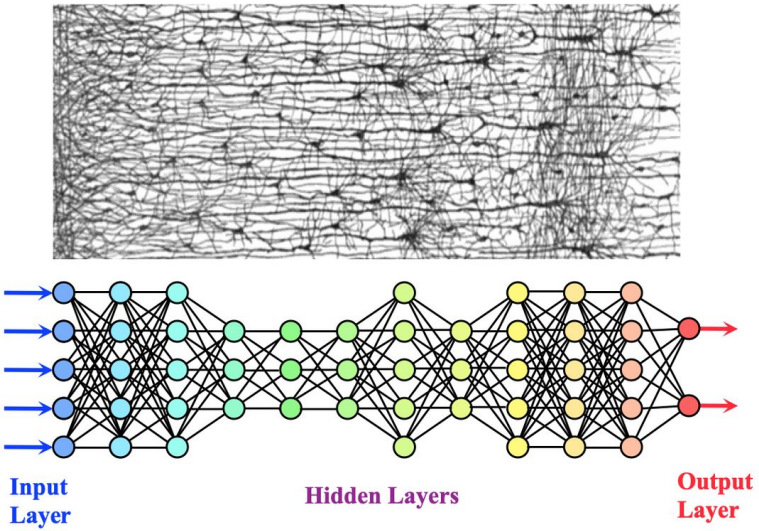
Key idea:

💡 AI does not need to think like humans to beat humans at certain problems.

The History of AI: The Deep Learning Revolution (2012)

Computers Learn From Data

- Deep learning became much more powerful.
- Neural networks improved dramatically.
- AI became better at recognizing images and speech.



Powered by:

- More data
- Faster computers
- Better algorithms

Key idea:
💡 AI started learning patterns instead of just following rules.

The History of AI: AI Masters Complex Games (2016)

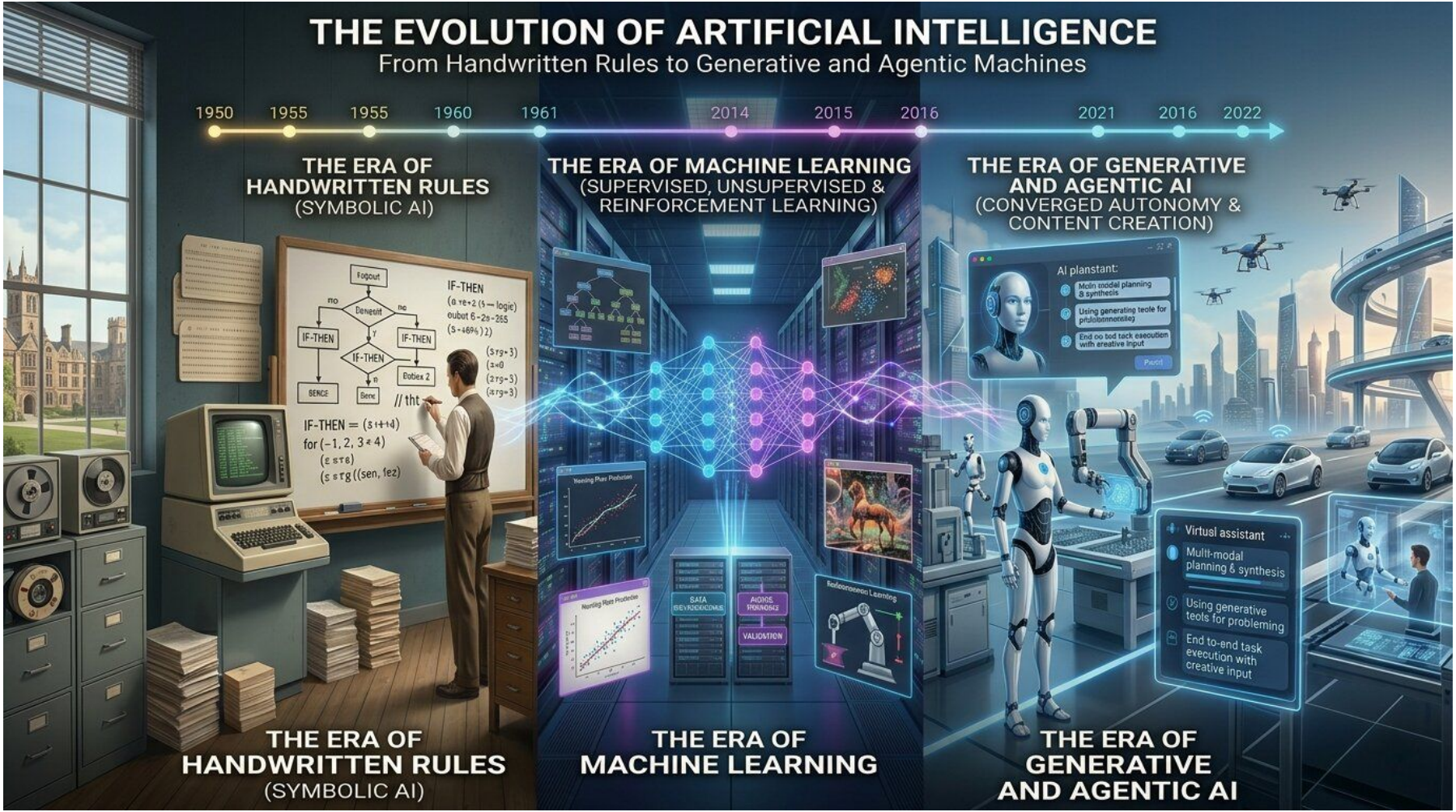
AlphaGo Defeats a Go Champion

- Google's AlphaGo defeated one of the world's best Go players.
- Go is much more complex than chess.
- AI used deep learning to discover strategies humans had never created.



Key idea:

💡 AI can discover solutions humans may not expect.


THE EVOLUTION OF ARTIFICIAL INTELLIGENCE
From Handwritten Rules to Generative and Agentic Machines
1950
1955
1955
1960
1961
2014
2015
2016
2021
2016
2022
THE ERA OF HANDWRITTEN RULES
(SYMBOLIC AI)
THE ERA OF MACHINE LEARNING
(SUPERVISED, UNSUPERVISED & REINFORCEMENT LEARNING)
THE ERA OF GENERATIVE AND AGENTIC AI
(CONVERGED AUTONOMY & CONTENT CREATION)
IF-THEN
IF-THEN
IF-THEN
IF-THEN
IF-THEN = (s i+i4)
for (−1, 2, 3 ≠ 4)
Virtual assistant
Multi-modal planning & synthesis
Using generative teols for problemimg
End to-end task execution with creative input
THE ERA OF HANDWRITTEN RULES
(SYMBOLIC AI)
THE ERA OF MACHINE LEARNING
THE ERA OF GENERATIVE AND AGENTIC AI

The History of AI: Today & Future

Where Is AI Going?

AI is helping develop:

- 🤖 Smarter robots
- 🚗 Autonomous vehicles
- 🏥 Medical discoveries
- 🌎 Climate solutions
- 🎓 Personalized education



Discussion Question:

What is one thing you hope AI can help humans do in the future?

Humans vs. AI

Humans learn from experience. AI learns from data.

How Do Humans and AI Learn?



Humans Learn From:

- Experiences
- Mistakes
- Observations
- Practice

Example:
A child learns what a dog looks like after seeing many dogs.

AI Learns From:

- Data
- Examples

Example:
An AI learns what a dog looks like after analyzing thousands of pictures.

What Is Data?

Data: The Information AI Learns From

AI learns from many types of data:

🖼️ Images

- Photos
- Videos
- Drawings

📝 Text

- Books
- Articles
- Messages

🎵 Audio

- Music
- Speech
- Sounds

🔢 Numbers

- Prices
- Weather
- Measurements



Training AI

How Does AI Learn?

Step 1: Give AI Examples

🐶 1,000 Dog Pictures

🐱 1,000 Cat Pictures

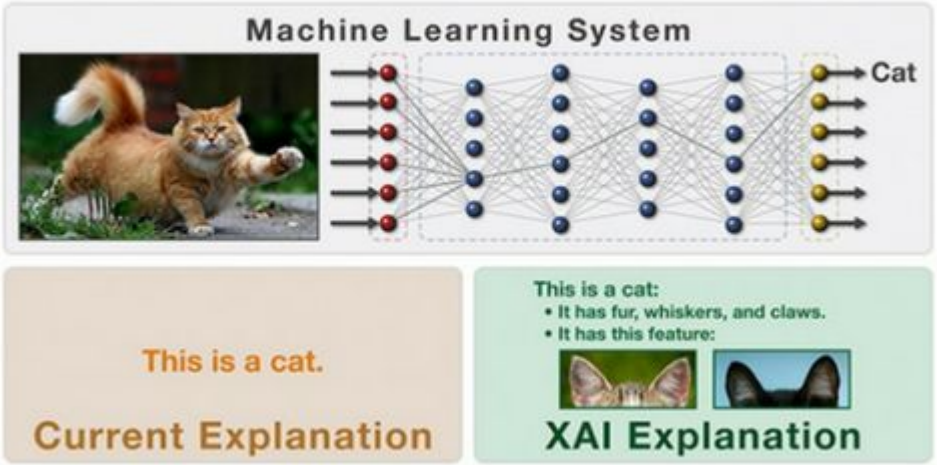
Step 2: AI Finds Patterns

AI learns:

- Shapes
- Colors
- Features
- Differences

Step 3: AI Creates a Model

A model that can recognize new images.



Testing AI

Can AI Recognize Something New?

New Picture:

❓ Is this a dog or a cat?

AI analyzes patterns:

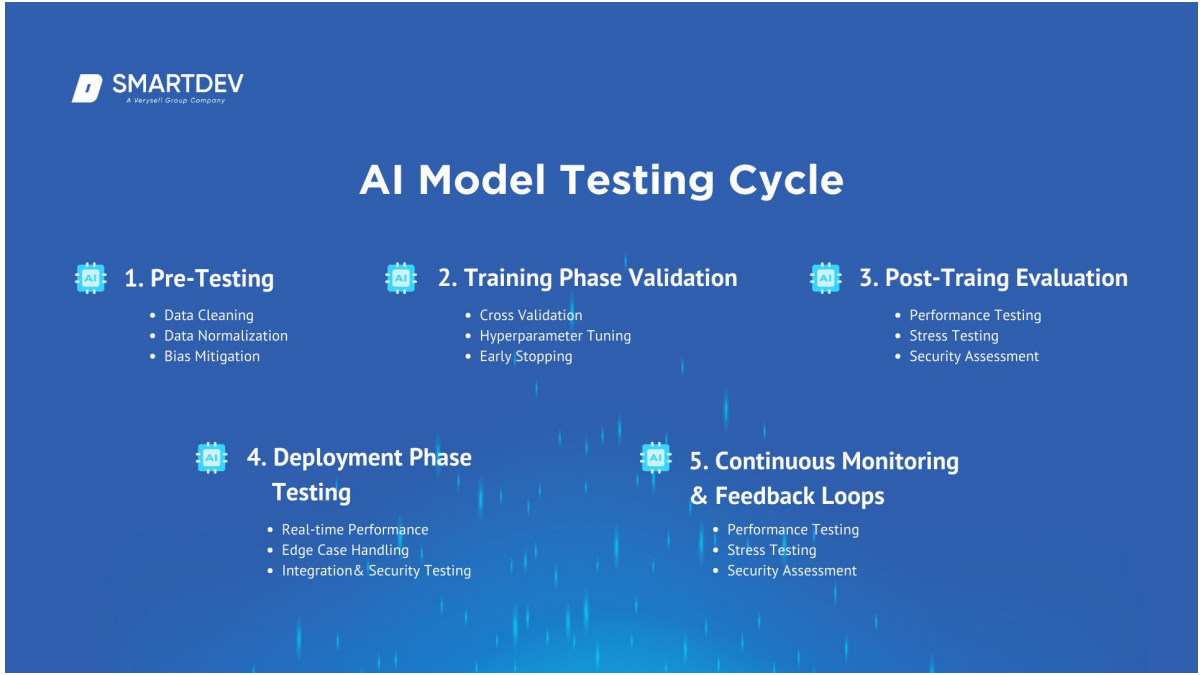
- Ears
- Shape
- Fur
- Face

Prediction:

🐶 Dog

or

🐱 Cat



Machine Learning

What Is Machine Learning?

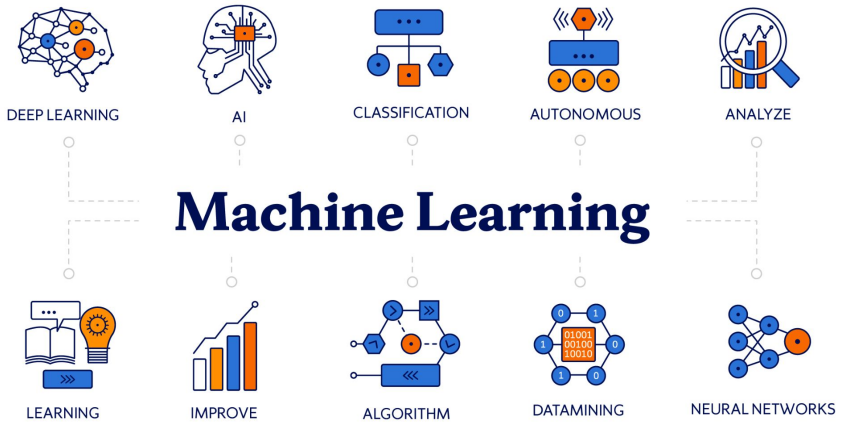
Machine Learning = Teaching computers to learn from examples instead of giving every rule.

Traditional Programming:

Human writes rules
Computer follows rules

Machine Learning:

Give examples
Computer finds patterns



Syracuse University
School of Information Studies

Deep Learning

Deep Learning: Learning With Many Layers

Deep Learning uses:

🧠 **Artificial Neural Networks**

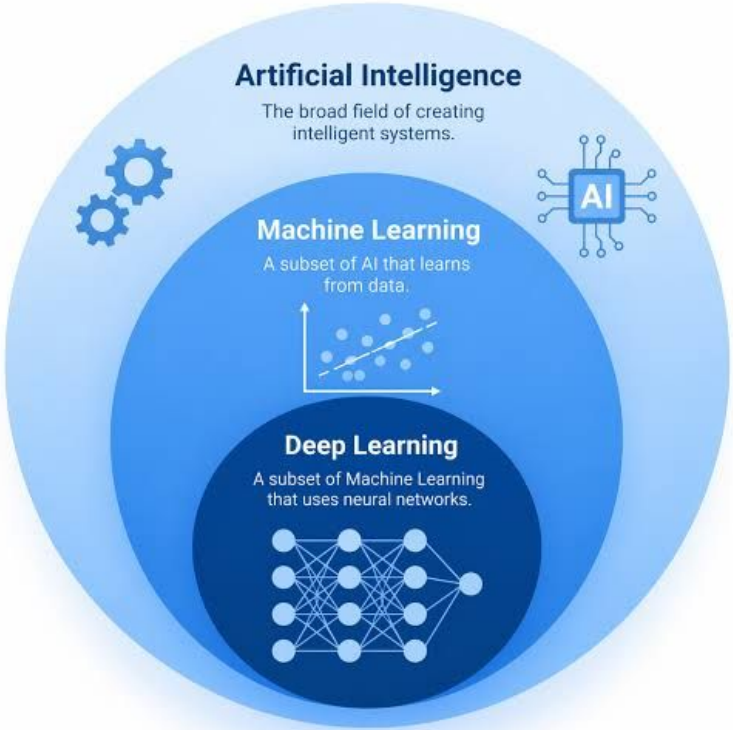
Inspired by the human brain.

Neural networks contain:

- Many connected "neurons"
- Multiple layers
- Pattern recognition

Deep Learning helps AI:

- Recognize faces
- Understand speech
- Generate images
- Translate languages



Large Language Models (LLMs)

How Does ChatGPT Work?

ChatGPT is a:

Large Language Model (LLM)

It learns patterns from huge amounts of text.

Example:

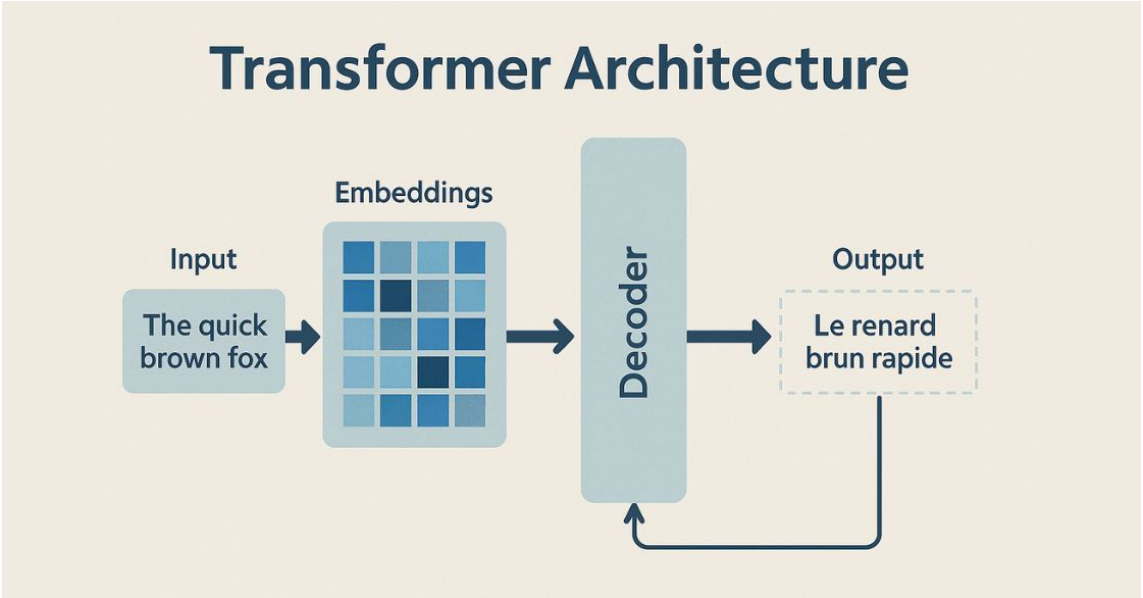
"I went to the ____"

What comes next?

Possible answers:

- School
- Store
- Park

ChatGPT predicts the most likely next word.



Why do LLMs Sound so Smart?

AI Learns Language Patterns

ChatGPT has learned from:

📚 Books

🌐 Websites

📝 Articles

💬 Conversations

It learns:

- How words connect
- How sentences are formed
- How ideas relate



AI in Healthcare

AI Helps Doctors Save Lives



AI is helping doctors:

- Detect diseases earlier
- Analyze medical images
- Find patterns in patient data

Examples:

🩻 **X-rays**

- AI can find signs of diseases in X-ray images.

🔬 **Cancer Detection**

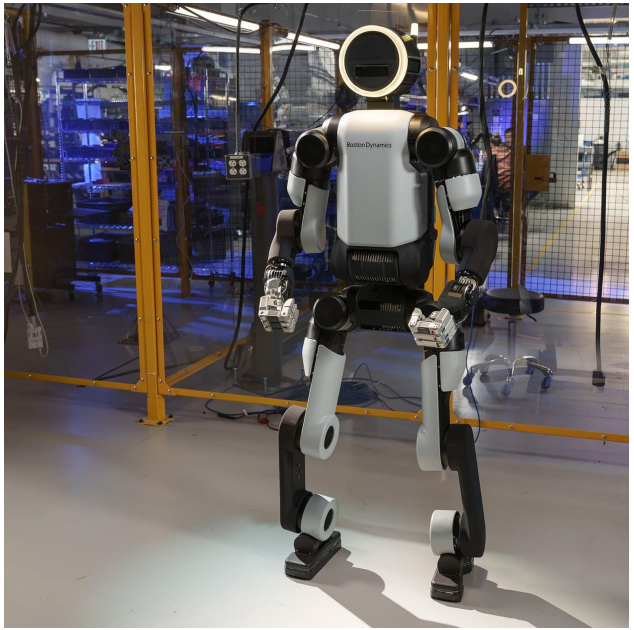
- AI can help identify cancer cells and tumors.

AI Robots

AI Gives Robots Intelligence

AI allows robots to:

- Understand their environment
- Make decisions
- Complete difficult tasks



Examples:

Boston Dynamics Robots

- Walk and balance like humans
- Carry objects
- Work in dangerous environments

Key Idea:

AI helps robots interact with the real world.

AI Self-Driving Cars

Teaching Cars to Drive

Self-driving cars use AI to understand:

- Roads
- Traffic signs
- Other vehicles
- Pedestrians





Examples:

Tesla

- Uses AI to assist drivers.

Waymo

- Uses AI to test fully autonomous vehicles.

AI in Education

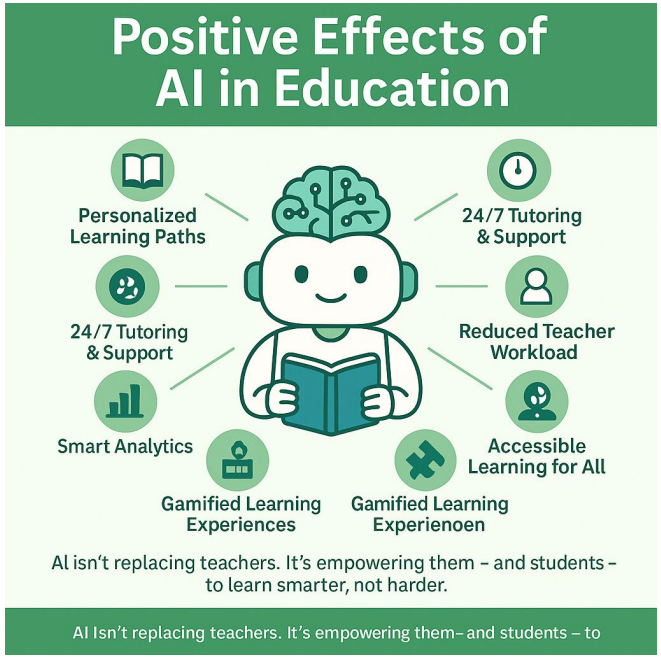
AI as a Learning Assistant

AI can help students:

- Explain difficult concepts
- Practice problems
- Give feedback
- Create personalized lessons

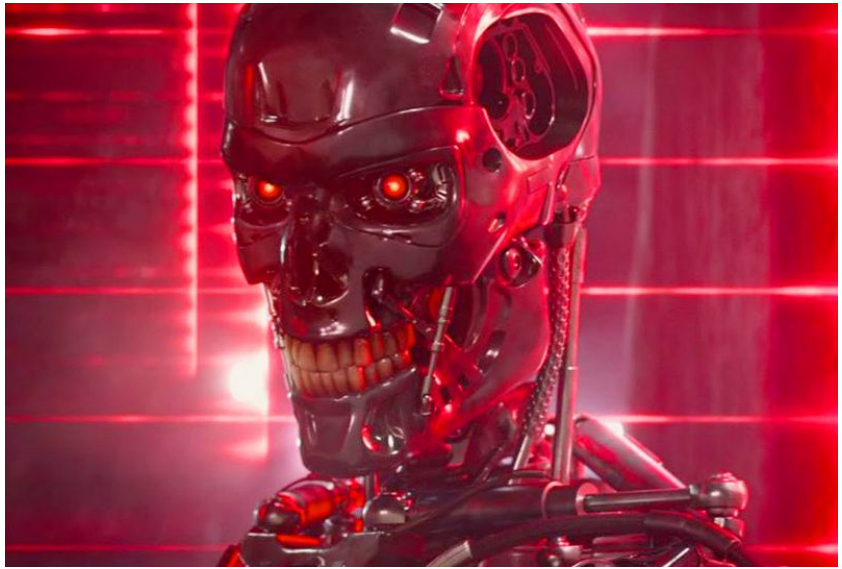
Examples:

📚 AI tutors
✏️ Writing assistants
🧮 Math helpers



AI Ethics

Should We Always Trust AI?



AI can do amazing things...

But we also need to ask:

- Is AI fair?
- Is AI safe?
- Should AI make certain decisions?
- Who is responsible when AI makes mistakes?

AI Bias

⚖️ AI Can Be Unfair

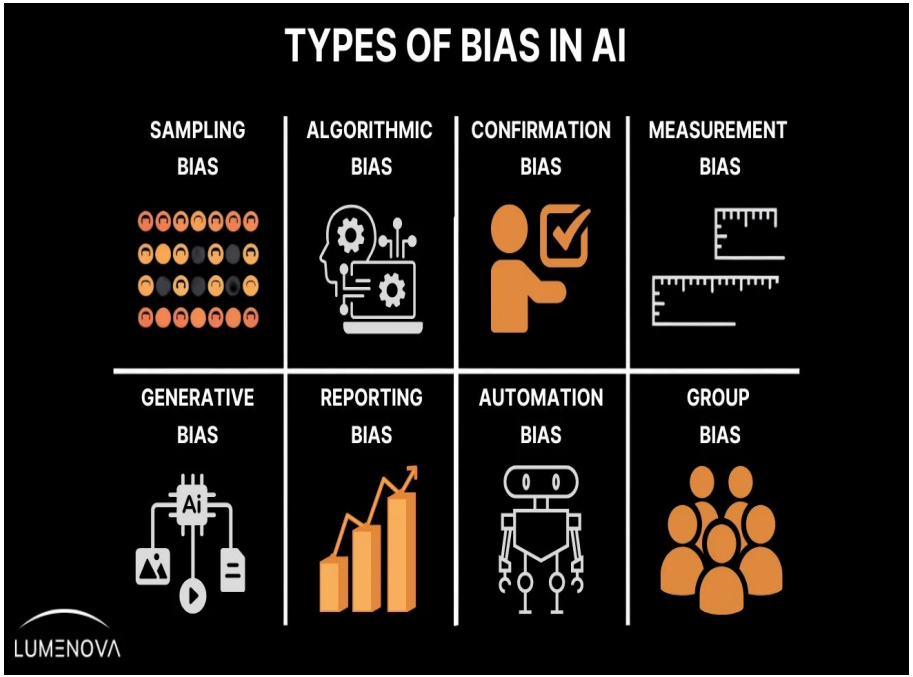
AI learns from data.

If the data contains mistakes or unfair patterns...

➡️ AI may learn those biases.

Examples:

- Face recognition mistakes
- Unfair recommendations
- Incorrect predictions



AI Hallucinations

When AI Makes Things Up

Sometimes AI creates information that sounds correct but is false.

This is called:

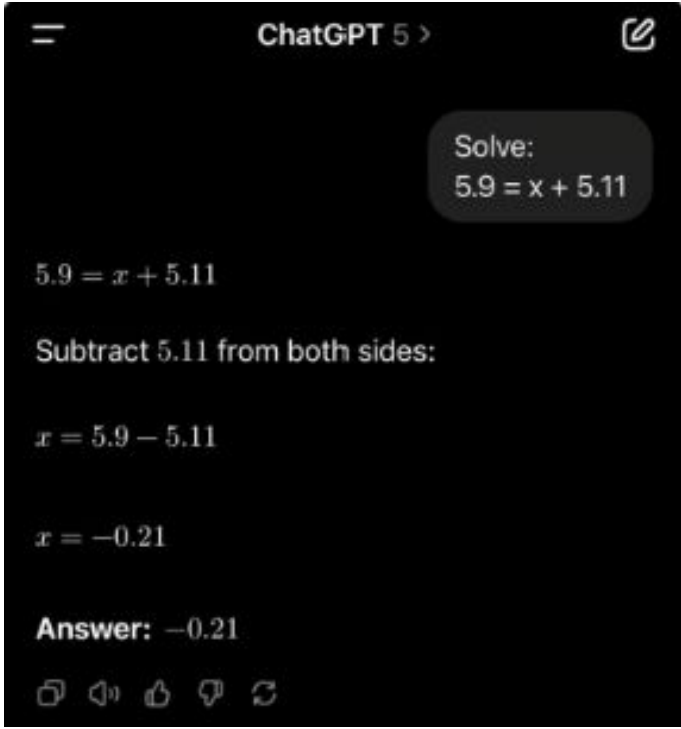
"AI Hallucination"

Examples:

- Fake facts
- Made-up sources
- Incorrect explanations

Key Idea:

Always check important information from AI.



A Preview of Next Week!

Have you wondered:

- **How many points will I score in my next basketball/soccer game?**
- **How much ice cream will a store sell on a hot summer day?**
- **How much food will a pet need based on its size**


COME TO THE AI
WORKSHOP NEXT WEEK
on August 1!!!!!!!!